Ayush Daga

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Summary

Data Analyst with experience in SQL, Python, and business intelligence, delivering insights from large datasets in cloud-based environments. Skilled in Power BI, ETL pipelines, and data validation, with hands-on experience in Databricks and PySpark for distributed data processing. Proven ability to analyze datasets, build dashboards, and support stakeholder decision-making in fast-paced, collaborative settings.

Education

Master of Science in Data Analytics **Aug 2024 - Present**
George Mason University, Virginia - **GPA: 4.0/4.0**
Relevant Coursework:
Machine Learning, Statistics, Principles of Data Mining, Big Data Essentials, Operations Research, Health Data Integration

Bachelor of Technology in Computer Science and Engineering **Aug 2018 - May 2022**
R.V College Of Engineering, Bangalore

Technical Skills

Languages & Frameworks: Python, R, SQL, Javascript, React, Angular, C/C++
Cloud and Big Data AWS (S3, EC2, Lambda, RDS, SageMaker), Databricks, PySpark, Spark MLlib
Data Visualization and BI Tools: Power BI, Tableau, Apache Superset, GaiaViz, Excel, matplotlib, seaborn ,Potly, ggplot2
Databases and Data Management: PostgreSQL, MySQL, SQL Optimization, ETL Pipelines, Jupyter, Git/GitHub
Machine Learning and Analytics: Statistical Analysis, Predictive Modeling, A/B Testing, EDA, Data Wrangling, KPI Reporting,Geospatial Analysis

Work Experience

Data Analyst — Data Agency and Security Lab, USA **Sept 2025 - Present**
- Analyze datasets from 61+ healthcare platforms using Python and Pandas, identifying security patterns across 5+ risk categories
- Build automated data pipelines using Python and PySpark concepts to process 133 mobile applications, improving data processing efficiency by 30%
- Generate structured analytical reports summarizing findings for research publication, supporting stakeholder understanding of risk patterns across 3+ research outputs
- Design data collection instruments and survey workflows aligned with IRB protocols for a multiphase user behavior study.

Data Analyst — Tech Mahindra, India **April 2022 - Sept 2022**
- Designed and deployed a MySQL–Apache Superset data visualization platform supporting insights for 1,000+ users.
- Automated Excel–MySQL pipelines and data cleaning, improving retrieval speed by 40% and ensuring 99% data integrity.
- Applied time series analysis and anomaly detection to cut anomalies by 25% and increased dashboard adoption by 25%.
- Developed an internal project management dashboard using MySQL and Apache, improving oversight efficiency by 30%.
- Optimized visualization workflows by minimizing data preprocessing, reducing dashboard development time by 40%.

Projects

3D Data Fusion for Real-Time Situational Awareness (GaiaViz LLC) — USA **Jan 2026 - Present**
- Building a Python ETL pipeline to ingest and transform real-time traffic data from VDOT's SmarterRoads REST API
- Analyzing detector-level metrics (vehicle count, speed, occupancy) across 3 major highway corridors in Northern Virginia.
- Creating geospatial visualizations using GaiaViz to map traffic KPIs, enabling data-driven transportation insights.

Handwritten Digit Classification Using MNIST Dataset — USA **Aug 2025 - Dec 2025**
- Built an end-to-end ML pipeline in Python for digit classification, covering data preprocessing, model training, and evaluation.
- Compared 4 models (Logistic Regression, Random Forest, MLP, CNN) achieving 99% accuracy with CNNs and full benchmarking.
- Conducted error analysis using confusion matrices to identify misclassification patterns across ML and deep learning approaches.

Wildfire Risk Prediction and Intensity Assessment — USA **Jan 2025 - May 2025**
- Built a scalable wildfire risk prediction pipeline using PySpark and Spark MLlib on Databricks, processing 150K+ records.
- Engineered 10+ domain-specific features and generated 60K+ synthetic samples for model robustness.
- Achieved 91% accuracy and 0.73 AUC-ROC using Random Forest; performed full data pipeline development and tuning.

Multi-Tool Framework for ARV Pricing and Supply Chain Analysis — USA **Aug 2024 - Dec 2024**
- Built an end-to-end data analytics pipeline using Python, MySQL, Excel, and Apache Superset, processing 50K+ records across 10+ countries to analyze ARV pricing and supply chain costs.
- Performed data cleaning, SQL analysis, and EDA, identifying 20% pricing variation drivers and reducing inconsistencies by 25%.
- Developed interactive dashboards and applied trend forecasting, improving reporting efficiency by 40% and enabling faster, data-driven supply chain decisions.